

TP1 – ECUACIONES DIFERENCIALES, 1RA. PARTE

- 01)
- a) orden 3 – grado 1. $y'' + y' + y = 0$
 - b) orden 3 – grado 1. $y'' + (y')^2 = 2y(y')^2$
 - c) orden 3 – grado 1
 - d) orden 2 – no hay grado
 - e) $3x dy = y dx \Rightarrow \frac{dy}{dx} 3x = y \Rightarrow y' 3x = y$
orden 1 – grado 1 – lineal
 - f) orden 2 – grado 1 – lineal

- 02)
- a) $y = e^{-x} + x - 1 \Rightarrow y' = -e^{-x} + 1$
 - $y(0) = e^{-0} + 0 - 1 = 1 - 1 = 0$ verifica $y(0) = 0$
 - $y' + y = x \Rightarrow (-e^{-x} + 1) + (e^{-x} + x - 1) = x \Leftrightarrow 0 = 0$
 - $\Leftrightarrow -e^{-x} + e^{-x} + x - 1 = x \Leftrightarrow 0 = 0$ verifica

$-x^{-1/2}$	HOJA * FECHA
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2). b) $y' = -\frac{1}{x} \sqrt{x} + \frac{1}{2\sqrt{x}} (2 - \ln(x))$

$$y'' = \frac{1}{2} x^{-3/2} + \frac{1}{2} \left[\left(-\frac{1}{2} x^{-3/2} \right) (2 - \ln x) - \frac{1}{x} \frac{1}{\sqrt{x}} \right] =$$

$$= \frac{1}{2} x^{-3/2} - \frac{1}{4} x^{-3/2} \cdot 2 + \frac{1}{4} x^{-3/2} \ln x - \frac{1}{2} x^{-3/2} =$$

$$= \frac{1}{4} x^{-3/2} \ln x - \frac{1}{2} x^{-3/2}$$

$$4x^2 y'' + y = 4x^2 \left(\frac{1}{4} x^{-3/2} \ln x - \frac{1}{2} x^{-3/2} \right) + (2 - \ln x) \sqrt{x} =$$

$$= x^{1/2} \ln x - 2x^{1/2} + 2\sqrt{x} - \sqrt{x} \ln x = 0 \quad \text{verifica la EDO}$$

- $x > 0$ para aplicar $\sqrt{x}$ y $\ln x$
- recta tangente en $(1, 2)$ es $y = 2$

$$y(1) = 2 \rightarrow y(1) = (2 - \ln 1) \sqrt{1} = (2 - 0) = 2$$

$$y'(1) = 0 \rightarrow y'(1) = -\frac{1}{1} \sqrt{1} + \frac{1}{2\sqrt{1}} (2 - \ln 1) = -1 + \frac{1}{2} \cdot 2 = 0$$

2)c) Verifique que $y^2 = C_1 x + C_2$ es SG de
 $y \cdot (y')^2 + y^2 y'' = 0$. Halle la SP que en $(1, y_0)$
tiene recta tangente de ecuación $y = 2x - 1$

Derivamos para sustituir en la ED y verificar
que se cumple la igualdad

$$y^2 = C_1 x + C_2 \Rightarrow 2y y' = C_1 \Rightarrow y' = \frac{C_1}{2} \cdot \frac{1}{y}$$

$$y' = \frac{C_1}{2} \cdot \frac{1}{y} \Rightarrow y'' = \frac{C_1}{2} \left(-\frac{1}{y^2} \right) \cdot y' \Rightarrow y'' = -\frac{C_1 y'}{2y^2} \Rightarrow$$

$$\Rightarrow y'' = \frac{C_1 \left(\frac{C_1}{2} \cdot \frac{1}{y} \right)}{2y^2} \Rightarrow y'' = -\frac{C_1^2}{4} \cdot \frac{1}{y^3}$$

$$\text{Reemplazamos: } y \cdot \left(\frac{C_1}{2} \cdot \frac{1}{y} \right)^2 + y^2 \left(-\frac{C_1^2}{4} \cdot \frac{1}{y^3} \right) =$$

$$= y \cdot \frac{C_1^2}{4} \cdot \frac{1}{y^2} - \frac{C_1^2}{4} \cdot \frac{1}{y} = 0 \quad \text{verifica la igualdad}$$

Demostremos que $y^2 = C_1 x + C_2$ es solución.

Además es SG porque la expresión tiene 2 constantes esenciales y arbitrarias (C_1 y C_2),

en concordancia con el orden de la ED
(es una ED de 2º orden)

$$SG \rightarrow y^2 = c_1 x + c_2$$

Hallamos la SP que en $(1, y_0)$ tiene recta tangente $y = 2x - 1$

$$y'(1) = 2 \quad (\text{pendiente de la recta})$$

$$y(1) = 2 \cdot 1 - 1 = 1 \quad (\text{la recta y la curva comparten el punto de tangencia})$$

$$y(1) = 1 \rightarrow 1^2 = c_1 \cdot 1 + c_2 \Rightarrow c_1 + c_2 = 1$$

$$y'(1) = 2 \rightarrow 2 = \frac{c_1}{2} \cdot \frac{1}{y} \Rightarrow 2 = \frac{c_1}{2} \cdot \frac{1}{1} \Rightarrow c_1 = 4$$

$$\begin{cases} c_1 + c_2 = 1 \\ c_1 = 4 \end{cases} \Rightarrow c_2 = -3$$

SP pedida:

$$y^2 = 4x - 3$$

$$d) \begin{cases} y = x \\ y' = 1 \\ y'' = 0 \end{cases} \Rightarrow y(y')^2 + y^2 y'' = x \cdot 1^2 + x^2 \cdot 0 = x$$

verifica

2)e) Verifique que:

$$\left. \begin{array}{l} y = cx + c^2 \text{ es SG} \\ y = -\frac{x^2}{4} \text{ es SS} \end{array} \right\} \text{ de } y = xy' + (y')^2$$

Halle las soluciones que pasan por $(2, -1)$

• Verificamos que $y = cx + c^2$ es solución

$y' = c$ Reemplazamos en la ED:

$$\underbrace{cx + c^2}_y = x \cdot \underbrace{c}_{y'} + \underbrace{c^2}_{(y')^2} \quad \text{Verifica la igualdad}$$

$y = cx + c^2$ es SG porque la expresión tiene una constante esencial y arbitraria, ya que la ED es de 1° orden

• Verificamos que $y = -\frac{x^2}{4}$ es solución

$y' = -\frac{2x}{4} = -\frac{x}{2}$ Reemplazamos en la ED:

$$\underbrace{-\frac{x^2}{4}}_y = x \cdot \underbrace{\left(-\frac{x}{2}\right)}_{y'} + \underbrace{\left(-\frac{x}{2}\right)^2}_{(y')^2}$$

$$-\frac{x^2}{4} = -\frac{x^2}{2} + \frac{x^2}{4} \Rightarrow -\frac{x^2}{4} = -\frac{x^2}{4} \quad \text{Verifica}$$

$y = -\frac{x^2}{4}$ es SS porque no se desprende de la SG (la SG es un haz de rectas y la SS es una parábola)

- Hallamos las soluciones que pasan por $(2, -1)$

$$SG \rightarrow y = cx + c^2 \quad \text{tomando que } y(2) = -1$$

$$-1 = 2 \cdot c + c^2 \Rightarrow c^2 + 2c + 1 = 0 \Rightarrow$$

$$\Rightarrow (c+1)^2 = 0 \Rightarrow c+1=0 \Rightarrow c = -1$$

$$\text{La SP } y = -1 \cdot x + (-1)^2$$

$$\boxed{y = -x + 1 \quad \text{pasa por } (2, -1)}$$

- Verificamos si la SS pasa por $(2, -1)$

$$SS: y = -\frac{x^2}{4} \quad \therefore \quad -1 = -\frac{2^2}{4} \quad \text{verifica}$$

$$\boxed{y = -\frac{x^2}{4} \quad \text{pasa por } (2, -1)}$$

f) $x^2 + 4y^2 = C \Rightarrow 2x + 8y y' = 0 \Rightarrow y' = -\frac{2x}{8y} = -\frac{x}{4y}$

- $4y y' = -x \Rightarrow 4y \left(-\frac{x}{4y}\right) = -x \Rightarrow -x = -x \quad \text{verifica}$
- $y(-2) = 1 \Rightarrow 4 + 4 \cdot 1 = C \Rightarrow C = 8$
 SP: $x^2 + 4y^2 = 8$

3)a) Hallar la ED correspondiente a la sig. familia de curvas $y^2 = 4ax$

$$y^2 = 4ax \quad (1 \text{ de esencial} \rightarrow 1^\circ \text{ orden})$$

$$\text{Derivamos MAM: } 2y y' = 4a \Rightarrow y y' = 2a$$

Para eliminar la constante a , despejamos de la ecuación de la familia de curvas y sustituimos en la ED

$$\left[\begin{array}{l} y^2 = 4ax \Rightarrow a = \frac{y^2}{4x} \\ y y' = 2a \end{array} \right] \Rightarrow y y' = \frac{2}{4x} \frac{y^2}{1} \Rightarrow$$

$$\Rightarrow \boxed{y' = \frac{y}{2x}}$$

$$3)b) x^2 + y^2 = r^2 \Rightarrow 2x + 2yy' = 0 \Rightarrow \boxed{x + yy' = 0}$$

$$3) c) \quad y = \text{sen}(ax+b)$$

2^{da} es esencial
↓
2º orden

$$y = \text{sen}(ax+b) \quad (1)$$

$$y' = a \cos(ax+b) \quad (2)$$

$$y'' = -a^2 \text{sen}(ax+b) \quad (3)$$

$$\text{De } (1) \text{ y } (3) \rightarrow y'' = -a^2 y \quad (4)$$

$$\text{De } (2) \quad a = \frac{y'}{\cos(ax+b)} \Rightarrow -a^2 = -\frac{(y')^2}{\cos^2(ax+b)} \quad (5)$$

siendo $\cos^2 \alpha = 1 - \text{sen}^2 \alpha$, y utilizando la ecuación (1) resulta:

$$\cos^2(ax+b) = 1 - \underbrace{\text{sen}^2(ax+b)}_{y^2} = 1 - y^2$$

Reemplazando la expresión obtenida en (5):

$$-a^2 = -\frac{(y')^2}{1-y^2} \quad (6)$$

$$\text{De } (4) \text{ y } (6) : \quad y'' = -\frac{(y')^2}{1-y^2} \cdot y$$

ED del
hoz de curros

$$y'' = \frac{(y')^2}{y^2-1} \cdot y$$

$$1) \textcircled{1} y = ae^x + bxe^x$$

$$\textcircled{2} y' = ae^x + be^x + bxe^x$$

$$\textcircled{3} y'' = ae^x + be^x + be^x + bxe^x = ae^x + 2be^x + bxe^x$$

$$\begin{aligned} \textcircled{2} - \textcircled{3} &\Rightarrow y' - y'' = \cancel{ae^x} + \cancel{be^x} + \cancel{bxe^x} - \cancel{ae^x} - 2\cancel{be^x} - \cancel{bxe^x} \\ &\Rightarrow y' - y'' = -be^x \Rightarrow b = \frac{y' - y''}{e^x} = \frac{y'' - y'}{e^x} \textcircled{*} \end{aligned}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow y + y' = ae^x + bxe^x + ae^x + be^x + bxe^x \Rightarrow$$

$$\Rightarrow y + y' = 2ae^x + 2bxe^x + be^x \Rightarrow$$

$$\Rightarrow \frac{y + y' - 2bxe^x - be^x}{2e^x} = a$$

$$\Rightarrow a = \frac{y + y' - 2 \frac{y'' - y'}{e^x} \times e^x - \frac{y'' - y'}{e^x} e^x}{2e^x} \Rightarrow$$

$$\Rightarrow a = \frac{y + y' - 2y'' + 2y' - y'' + y'}{2e^x}$$

$$\Rightarrow a = \frac{y + 2y' - 2y'' + 2y' - y''}{2e^x} \textcircled{*}$$

Reemplazando $\textcircled{*}$ en $\textcircled{1}$:

$$y = \frac{y + 2y' - 2y'' + 2y' - y''}{2e^x} \times e^x + \frac{y'' - y'}{e^x} \times e^x \Rightarrow$$

$$\Rightarrow y = \frac{y}{2} + y' - \cancel{y''} + y' - \frac{y''}{2} + \cancel{y''} - y' \Rightarrow$$

$$\Rightarrow y' - \frac{y''}{2} + \frac{y}{2} - y = 0 \Rightarrow -\frac{y''}{2} + y' - \frac{y}{2} = 0 \Rightarrow$$

$$\Rightarrow -y'' + 2y' - y = 0$$

3)e) $y = c_1 x + \frac{c_2}{x} + c_3$ 3 cte $\rightarrow 3^\circ$ orden

$$y' = c_1 - \frac{c_2}{x^2}$$

$$y'' = -(-2)c_2 x^{-3} = 2c_2 x^{-3} \quad (1)$$

$$y''' = -6c_2 x^{-4} \quad (2)$$

De (1) $\rightarrow c_2 = \frac{y''}{2} \cdot x^3$ Reemplazando en (2):

$$y''' = -6 \frac{y''}{2} \cdot x^3 \cdot x^{-4} \Rightarrow y''' = -3 \frac{y''}{x} \Rightarrow$$

$$\boxed{\Rightarrow y''' x = -3y''}$$

f) $y = b a^x \quad (1)$

$$y' = b a^x \ln a \quad (2)$$

$$y'' = b \ln a a^x \ln a \Rightarrow y'' = b \ln^2 a a^x \quad (3)$$

De (1) $\rightarrow b = y a^{-x}$

Reempl en (2): $y' = y a^{-x} a^x \ln a \Rightarrow y' = y \ln a \Rightarrow$

$$\Rightarrow \frac{y'}{y} = \ln a \Rightarrow e^{y'/y} = a$$

Reempl en (3) $\rightarrow y'' = y \cancel{[e^{y'/y}]^{-x}} \ln^2 (e^{y'/y}) \cancel{[e^{y'/y}]^x}$

$$\Rightarrow y'' = y \ln^2 (e^{y'/y}) \Rightarrow y'' = y \left(\frac{y'}{y} \right)^2 \Rightarrow y'' = \frac{(y')^2}{y}$$

$$\Rightarrow y'' y = (y')^2$$

4)

a) $y = mx + b \Rightarrow -1 = m \cdot 1 + b \Rightarrow b = -1 - m$

$$y = mx - 1 - m \Rightarrow y = \overset{y}{m}(x-1) - 1 \Rightarrow \boxed{y'(x-1) = y+1}$$

4) b) Halle la ED de la familia de curvas hipérbolas con focos en el eje x , centro en el origen y semiejes a variable y
 $b = 1$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \wedge \quad b = 1 \Rightarrow \frac{x^2}{a^2} - y^2 = 1 \quad (1)$$

Derivemos HAM: $\frac{2x}{a^2} - 2y y' = 0 \quad (2)$

Debemos sustituir la constante en la ED

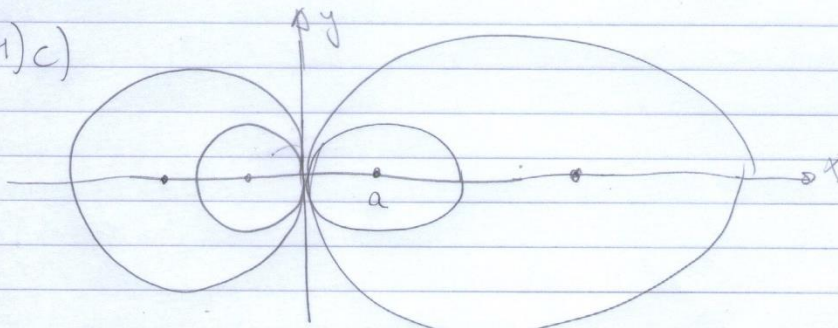
De (1): $\frac{x^2}{a^2} = 1 + y^2 \Rightarrow a^2 = \frac{x^2}{1 + y^2}$

Reemplazando la expresión anterior en (2)

$$\frac{2x}{\frac{x^2}{1+y^2}} - 2y y' = 0 \Rightarrow 2 \left[\frac{1+y^2}{x} - y y' \right] = 0 \Rightarrow$$

$$\Rightarrow \boxed{\frac{1+y^2}{x} - y y' = 0}$$

4) c)



$$(x-a)^2 + y^2 = \underbrace{r^2}_{a^2} \rightarrow a^2$$

$$(x-a)^2 + y^2 = a^2 \Rightarrow x^2 - 2ax + \cancel{a^2} + y^2 = \cancel{a^2} \Rightarrow$$

$$\Rightarrow x^2 - 2ax + y^2 = 0 \Rightarrow x^2 + y^2 = 2ax \quad (1)$$

Derivando MAM : $2x + 2yy' = 2a \Rightarrow x + yy' = a \quad (2)$

De (1) : $a = \frac{x^2 + y^2}{2x}$ reemplazo en (2)

$$x + yy' = \frac{x^2 + y^2}{2x} \Rightarrow 2x^2 + 2xyy' = x^2 + y^2 \Rightarrow$$

$$\boxed{\Rightarrow 2xyy' = y^2 - x^2}$$

$$5) [e^{x^3+2x} \cdot g]' = (e^{x^3+2x})' \cdot g'$$

$$\cancel{e^{x^3+2x}} \cdot (3x^2+2)g + \cancel{e^{x^3+2x}} g' = \cancel{e^{x^3+2x}} (3x^2+2) \cdot g'$$

$$(3x^2+2)g + g' = (3x^2+2)g' \Rightarrow$$

$$\Rightarrow (3x^2+2)g = (3x^2+1)g'$$

$$(3x^2+2)g = (3x^2+1) \frac{dg}{dx} \Rightarrow \int \frac{3x^2+2}{3x^2+1} dx = \int \frac{dg}{g} \Rightarrow$$

$$\Rightarrow \ln(kg) = \int \left(\frac{3x^2+1}{3x^2+1} + \frac{1}{3(x^2+1/3)} \right) dx \Rightarrow$$

$$\Rightarrow \ln(kg) = x + \frac{1}{3} \cdot \frac{1}{\sqrt{3}} \arctg\left(\frac{x}{\sqrt{3}}\right)$$

$$kg = e^{x + \frac{1}{3\sqrt{3}} \arctg(x/\sqrt{3})}$$

$$g = c e^{x + \frac{1}{3\sqrt{3}} \arctg(x/\sqrt{3})}$$

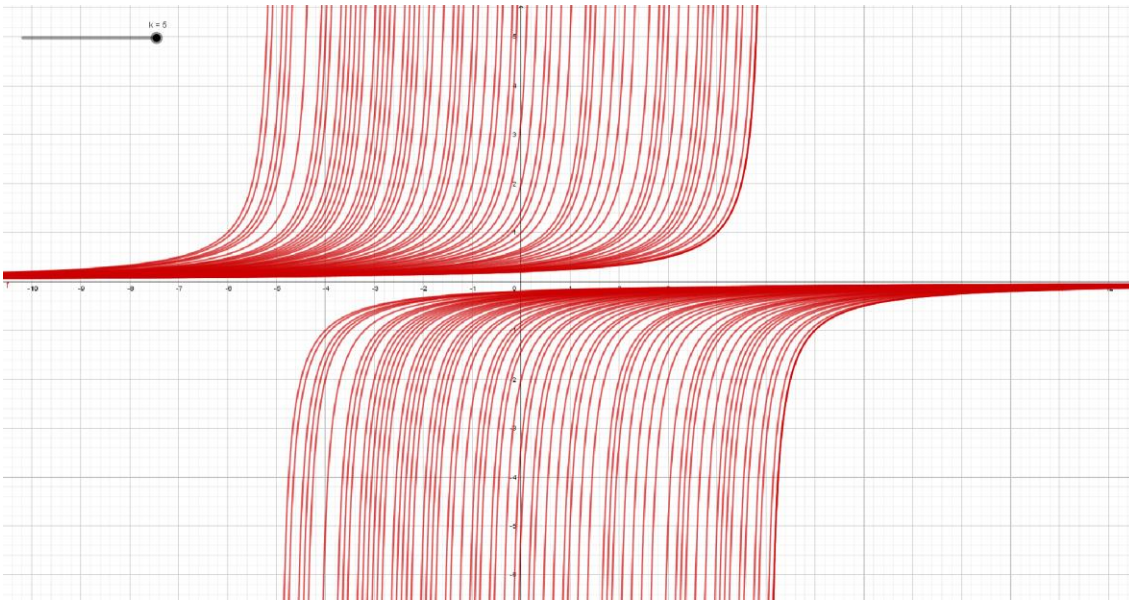
$$6) y = \frac{1}{c-x} \quad y' = \frac{1}{(c-x)^2} \cdot (-1) = \frac{1}{(c-x)^2} = y^2$$

verifica

$$y' = y^2 \Rightarrow \frac{dy}{dx} = y^2 \Rightarrow \int \frac{dy}{y^2} = \int dx \Rightarrow -y^{-1} = x + c \Rightarrow -\frac{1}{y} = x + c \Rightarrow -\frac{1}{x+c} = y \Rightarrow$$

$$\Rightarrow y = \frac{1}{-x-c} \Rightarrow y = \frac{1}{k-x} \text{ con } k = -c$$

Pero $y=0$ también es solución. La asíntota horizontal de todas las curvas es $y=0$, y resulta la solución singular de la ED (ver gráfico en próxima hoja)



7) a) $\frac{dy}{dx} = \frac{x^2+1}{2-y} \Rightarrow \int dy (2-y) = \int (x^2+1) dx \Rightarrow$

$$\Rightarrow \left[2y - \frac{y^2}{2} = \frac{x^3}{3} + x + k \right]$$

$y(-3)=4 \Rightarrow$
 $2 \cdot 4 - \frac{16}{2} = \frac{-27}{3} - 3 + k \Rightarrow$
 $\Rightarrow k=12$

7)b) Hallar la SG o SP según corresponda

$$x \frac{dy}{dx} - y = 2x^2 y$$

$$x \frac{dy}{dx} = 2x^2 y + y = y(2x^2 + 1)$$

$$x dy = y(2x^2 + 1) dx$$

$$\frac{dy}{y} = \frac{2x^2 + 1}{x} dx \Rightarrow \frac{dy}{y} = \left(2x + \frac{1}{x}\right) dx$$

$$\int \frac{dy}{y} = \int \left(2x + \frac{1}{x}\right) dx$$

$$\ln |y| + c = x^2 + \ln |x| \quad \text{con } c = \ln k :$$

$$\ln |y| + \ln k = x^2 + \ln |x|$$

$$\ln(k|y|) = x^2 + \ln |x| \quad \text{con } q = \pm k$$

$$\ln(qy) = \frac{x^2}{2} + \ln |x| \Rightarrow \ln(qy) - \ln |x| = \frac{x^2}{2} \Rightarrow$$

$$\Rightarrow \ln \left(\frac{qy}{|x|} \right) = \frac{x^2}{2} \quad \text{con } a = \pm q$$

$$\ln \left(a \frac{y}{x} \right) = \frac{x^2}{2} \Rightarrow e^{\frac{x^2}{2}} = \frac{a y}{x} \Rightarrow$$

$$\Rightarrow \frac{1}{a} x e^{\frac{x^2}{2}} = y \Rightarrow \boxed{y = b x e^{\frac{x^2}{2}}}$$

$$7)e) \frac{dy}{dx} = \frac{x}{\sqrt{x^2+9}} \Rightarrow \int dy = \int \frac{x}{\sqrt{x^2+9}} dx \Rightarrow$$

$$\Rightarrow y = \sqrt{x^2+9} + c$$

$$y(4) = 2 \Rightarrow 2 = \sqrt{16+9} + c \Rightarrow c = 2-5 = -3$$

$$\boxed{y = \sqrt{x^2+9} - 3}$$

$$7)d) x^2 dy = \frac{x^2+1}{3y^2+1} dx \Rightarrow \int dy (3y^2+1) = \int \frac{x^2+1}{x^2} dx \Rightarrow$$

$$\Rightarrow y^3 + y + k = \int \left(1 + \frac{1}{x^2}\right) dx \Rightarrow y^3 + y + k = x - \frac{1}{x}$$

$$y(1) = 2 \Rightarrow 8 + 2 + k = 1 - 1 \Rightarrow k = -10$$

$$\boxed{y^3 + y - 10 = x - \frac{1}{x}} \Rightarrow \begin{aligned} y^3 x + y x - 10 x &= x^2 - 1 \\ y^3 x + y x - x^2 - 10 x + 1 &= 0 \end{aligned}$$

$$7)c) \frac{dy}{dx} = 2x\sqrt{y-1} \Rightarrow \int \frac{dy}{\sqrt{y-1}} = \int 2x dx \Rightarrow$$

$$\Rightarrow \frac{(y-1)^{1/2}}{1/2} = x^2 + c$$

$$\Rightarrow 2\sqrt{y-1} = x^2 + c$$

$$7) f) y' = xy + x - 2y - 2 \text{ con } y(0) = 2$$

Primero hallamos la SG

$$y' = x(y+1) - 2(y+1) = (x-2)(y+1)$$

$$\frac{dy}{dx} = (x-2)(y+1) \Rightarrow \frac{1}{y+1} dy = (x-2) dx$$

$$\int \frac{1}{y+1} dy = \int (x-2) dx \Rightarrow \ln[k(y+1)] = \frac{x^2}{2} - 2x$$

$$\Rightarrow e^{x^2/2 - 2x} = k(y+1) \Rightarrow$$

$$\Rightarrow c e^{x^2/2 - 2x} = y+1 \Rightarrow y = c e^{x^2/2 - 2x} - 1$$

Reemplazamos para hallar la SP

$$y(0) = 2 \Rightarrow 2 = c e^{0-0} - 1 \Rightarrow 2 = c - 1 \Rightarrow c = 3$$

$$\text{La SP pedida es } y = 3e^{x^2/2 - 2x} - 1$$

$$8) \frac{dy}{dx} = \frac{x}{y} \Rightarrow \int dy y = \int x dx \Rightarrow \frac{y^2}{2} + k = \frac{x^2}{2} \Rightarrow$$

$$\Rightarrow \frac{y^2}{2} - \frac{x^2}{2} = -k \Rightarrow y^2 - x^2 = -2k \Rightarrow$$

$$\Rightarrow \boxed{y^2 - x^2 = c} \quad \text{hipérbolas equiláteras}$$

$$9) y' = xy \quad y \text{ pasa por } (0, -2)$$

$$\frac{dy}{dx} = xy \Rightarrow \int \frac{dy}{y} = \int dx x \Rightarrow \ln y = \frac{x^2}{2} + c$$

$$\Rightarrow e^{x^2/2 + c} = y \Rightarrow y = e^{x^2/2} \cdot f \quad (f = e^c)$$

$$-2 = e^0 \cdot f \Rightarrow f = -2 \rightarrow \boxed{y = -2e^{x^2/2}}$$

10) a) Halle la familia de curvas tales que su recta tangente en cada punto pase por (0,0)

Ecuación de un haz de rectas que pase por (0,0)

$$y = m x$$

↑ la pendiente está dada por y' porque es un haz de rectas tangentes a la familia de curvas

$$y = y' x \quad \text{ED de la familia de curvas}$$

Resolvamos la ED y su SG será la ecuación de la familia de curvas buscada

$$y = y' x \Rightarrow y = \frac{dy}{dx} x \Rightarrow \frac{dx}{x} = \frac{dy}{y} \Rightarrow \Rightarrow \ln(kx) = \ln y \Rightarrow kx = y$$

$$\boxed{y = kx}$$

10)

b) $y = k$
 $y' = 0 \Rightarrow \boxed{y = c}$

c) $y = mx + b$, $b = x + y \Rightarrow y = mx + x + y$

$\Rightarrow 0 = x(m+1)$ Si consideramos $x \neq 0$, entonces: $(m+1)=0$ siendo m la pendiente del haz de rectas tangentes. Dado que la pendiente de la recta tangente a una curva asociada a una función escalar es la derivada de dicha función, tenemos que:

$$m+1=0 \Rightarrow y'+1=0 \Rightarrow \frac{dy}{dx} = -1 \Rightarrow \int dy = \int -x dx \Rightarrow y = -x + c \quad (\text{con } x \neq 0)$$

10) d) Abscisa al origen: se refiere a la raíz de la recta

Ecuación general de un haz de rectas:

$$y = m x + b$$

$\uparrow$ y' por ser haz de rectas tangentes

Busco la raíz de $y = y' x + b$, luego:

$$y' x_0 + b = 0 \Rightarrow x_0 = -\frac{b}{y'} \quad (1)$$

Por dato del problema, la raíz es el triple de la abscisa del punto. Luego $x_0 = 3x$ (2)

De (1) y (2) resulta: $-\frac{b}{y'} = 3x \Rightarrow b = -3xy'$

Ecuación del haz de rectas: $y = y' x + b$ con $b = -3xy'$, entonces:

$$y = y' x - 3xy' \Rightarrow y = -2xy' \quad \text{ED del haz de curvas}$$

Resolvemos y la SG es la ecuación de la familia de curvas buscada

$$y = -2xy' \Rightarrow y = -2x \frac{dy}{dx} \Rightarrow \frac{dx}{x} = -\frac{2 dy}{y}$$

$$\Rightarrow \ln(kx) = -2 \ln y \Rightarrow \ln(kx) = \ln(y^{-2})$$

$$\Rightarrow kx = \frac{1}{y^2} \Rightarrow \boxed{xy^2 = C}$$

10)

e) $y_1 = ax + b$ (recta tge) pase por (x_0, y_0)
 $(x_0, ax_0 + b)$

Recta que pase por (x_0, y_0) y el origen de coord.

$$y_2 = mx \Rightarrow y_0 = mx_0 \Rightarrow m = \frac{y_0}{x_0}$$

Si $y_1 + y_2 = 0 \Rightarrow a = -\frac{1}{m} \Rightarrow a = -\frac{x_0}{y_0}$

$$y_1 = -\frac{x_0}{y_0} x + b$$

$$\begin{cases} y' = a \\ y' = -\frac{x}{y} \end{cases} \Rightarrow \frac{dy}{dx} = -\frac{x}{y} \Rightarrow \int dy y = \int -x dx \Rightarrow$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + c \Rightarrow y^2 + x^2 = 2c \Rightarrow x^2 + y^2 = k \quad (k > 0)$$

11) $y = kx^2 \Rightarrow y' = 2kx \Rightarrow y' = 2 \frac{y}{x}$

$y' = 2 \frac{y}{x}$ pendiente de la recta normal $\circ \circ$

$$-\frac{1}{y'} = 2 \frac{y}{x} \Rightarrow -\frac{dx}{dy} = \frac{2y}{x} \Rightarrow$$

$$\Rightarrow \int -dx x = \int 2y dy \Rightarrow -\frac{x^2}{2} = y^2 + c \Rightarrow$$

$$\Rightarrow y^2 + \frac{x^2}{2} = c \Rightarrow 2y^2 + x^2 = \phi$$

Pase por $(0, 1) \rightarrow 2 + 0 = \phi \Rightarrow \phi = 2$

$$2y^2 + x^2 = 2 \Rightarrow y^2 + \frac{x^2}{2} = 1$$

12) recta normal $y = -\frac{1}{y'} x + 5$

$$y - 5 = -\frac{dx}{dy} x \Rightarrow \int (y - 5) dy = \int -x dx \Rightarrow$$

$$\Rightarrow \frac{y^2}{2} - 5y = -\frac{x^2}{2} + c \Rightarrow \frac{y^2}{2} - 5y + \frac{x^2}{2} = c \Rightarrow$$

$$\Rightarrow y^2 - 10y + x^2 = k$$

13)a) $x y' - y - x^3 = 0$ Verifiquemos que sea lineal de 1° orden

$$x y' - y = x^3 \Rightarrow y' - \frac{1}{x} y = x^2$$

Substitución: $y = u \cdot v$ $y' = u'v + uv'$

Reemplazamos: $\underbrace{u'v + uv'}_{y'} - \frac{1}{x} \underbrace{uv}_y = x^2$

$$u'v + uv' - \frac{1}{x} uv = x^2$$

Sacamos factor común

$$u'v + u \underbrace{\left(v' - \frac{1}{x} v\right)}_{=0} = x^2 \quad (*)$$

$$v' - \frac{1}{x} v = 0 \Rightarrow \frac{dv}{dx} = \frac{v}{x} \Rightarrow \frac{dv}{v} = \frac{dx}{x} \Rightarrow$$

$$\Rightarrow \int \frac{dv}{v} = \int \frac{dx}{x} \Rightarrow \ln v = \ln x \Rightarrow \boxed{v = x}$$

Reemplazamos $v = x$ en (*)

$$u'x = x^2 \Rightarrow \frac{du}{dx} = x \Rightarrow du = x \, dx \Rightarrow$$

$$\Rightarrow \int du = \int x \, dx \Rightarrow \boxed{u = \frac{x^2}{2} + C}$$

Como la función $y = u \cdot v$, resulta

$$y = \left(\frac{x^2}{2} + C\right) x \Rightarrow \boxed{y = \frac{x^3}{2} + Cx} \quad \text{SG}$$

$$13) b) y' + y \cos x = \operatorname{sen} x \cos x$$

$$y = u v \Rightarrow y' = u' v + v' u$$

$$u' v + v' u + u v \cos x = \operatorname{sen} x \cos x$$

$$u' v + u (v' + v \cos x) = \operatorname{sen} x \cos x \quad (1)$$

$$\frac{dv}{dx} = -v \cos x \Rightarrow \int \frac{dv}{v} = \int -\cos x \, dx \Rightarrow$$

$$\Rightarrow \ln v = -\operatorname{sen} x \Rightarrow v = e^{-\operatorname{sen} x}$$

Reins en (1):

$$u' e^{-\operatorname{sen} x} = \operatorname{sen} x \cos x$$

$$\frac{du}{dx} = e^{\operatorname{sen} x} \operatorname{sen} x \cos x \Rightarrow \int du = \int e^{\operatorname{sen} x} \operatorname{sen} x \cos x \, dx = (A)$$

$$w = e^{\operatorname{sen} x} \Rightarrow \operatorname{sen} x = \ln w$$

$$dw = e^{\operatorname{sen} x} \cos x \, dx$$

$$(A) \Rightarrow u = \int \ln w \, dw \Rightarrow u = w - \ln w - w + c$$

$$\Rightarrow u = e^{\operatorname{sen} x} \operatorname{sen} x - e^{\operatorname{sen} x} + c$$

$$y = u v \Rightarrow y = (e^{\operatorname{sen} x} \operatorname{sen} x - e^{\operatorname{sen} x} + c) e^{-\operatorname{sen} x}$$

$$y = \operatorname{sen} x - 1 + c e^{-\operatorname{sen} x}$$

$$13) c) (x^2+4) y' - 3xy = x \Rightarrow y' - \frac{3x}{x^2+4} y = \frac{x}{x^2+4}$$

$$y = ur \Rightarrow y' = u' r + r' u$$

$$u' r + r' u - \frac{3x}{x^2+4} u r = \frac{x}{x^2+4}$$

$$u' r + u \left(r' - \frac{3x}{x^2+4} r \right) = \frac{x}{x^2+4}$$

$$r' - \frac{3x}{x^2+4} r = 0 \Rightarrow \frac{dr}{dx} = \frac{3x}{x^2+4} r \Rightarrow$$

$$\Rightarrow \int \frac{dr}{r} = \int \frac{3x}{x^2+4} dx \Rightarrow \ln r = 3 \frac{1}{2} \ln(x^2+4) \Rightarrow$$

$$\Rightarrow \ln r = \ln(x^2+4)^{3/2} \Rightarrow r = (x^2+4)^{3/2}$$

$$u' (x^2+4)^{3/2} = \frac{x}{x^2+4} \Rightarrow \frac{du}{dx} = \frac{x}{(x^2+4)(x^2+4)^{3/2}} \Rightarrow$$

$$\Rightarrow \int du = \int \frac{x}{(x^2+4)^{5/2}} dx \Rightarrow u = \frac{(x^2+4)^{-3/2}}{2} \cdot \left(-\frac{x}{3}\right) + C \Rightarrow$$

$$\Rightarrow u = -\frac{1}{3} \frac{1}{(x^2+4)^{3/2}} + C$$

$$y = ur = \left(-\frac{1}{3} \frac{1}{(x^2+4)^{3/2}} + C \right) (x^2+4)^{3/2} \Rightarrow$$

$$\Rightarrow y = -\frac{1}{3} + C (x^2+4)^{3/2}$$

$$(2^2)^{3/2} = 2^3 = 8$$

$$P_0 = (0, 1) \therefore 1 = -\frac{1}{3} + C \cdot 4^{3/2} = -\frac{1}{3} + C \cdot 8 \Rightarrow C = \frac{4}{3} \cdot \frac{1}{8} = \frac{1}{6}$$

$$y = -\frac{1}{3} + \frac{1}{6} (x^2+4)^{3/2}$$

$$13)d) \frac{dy}{dx} - \frac{2y}{x} = x^2 \operatorname{sen}(3x)$$

$$y' - \frac{2}{x} y = x^2 \operatorname{sen}(3x) \quad \text{Sustitución } y = u \cdot v$$

$$y' = u'v + uv'$$

$$\underbrace{u'v + uv'}_{\text{factor común}} - \frac{2}{x} uv = x^2 \operatorname{sen}(3x)$$

$$u'v + u \left(v' - \frac{2}{x} v \right) = x^2 \operatorname{sen}(3x) \quad (*)$$

$$v' - \frac{2}{x} v = 0 \Rightarrow \frac{dv}{dx} = \frac{2}{x} v \Rightarrow \frac{dv}{v} = \frac{2}{x} dx$$

$$\int \frac{dv}{v} = \int \frac{2}{x} dx \Rightarrow \ln v = 2 \ln x \Rightarrow$$

$$\Rightarrow \ln v = \ln x^2 \Rightarrow v = x^2 \quad \text{Reemplazamos en } (*)$$

$$u' x^2 = x^2 \operatorname{sen}(3x) \Rightarrow \frac{du}{dx} = \operatorname{sen}(3x) \Rightarrow$$

$$\Rightarrow du = \operatorname{sen}(3x) dx \Rightarrow \int du = \int \operatorname{sen}(3x) dx \Rightarrow$$

$$\Rightarrow u = -\frac{\cos(3x)}{3} + C$$

Como $y = u \cdot v$ resulta:

$$y = \left(-\frac{\cos(3x)}{3} + C \right) \cdot x^2$$

$$\boxed{y = -\frac{x^2 \cos(3x)}{3} + C x^2}$$

S.G.

14) Variables separables

$$y' = 6x - 2xy = 2x(3-y) \Rightarrow \frac{dy}{dx} = 2x(3-y) \Rightarrow$$

$$\Rightarrow \frac{dy}{3-y} = 2x dx \Rightarrow \int \frac{dy}{3-y} = \int 2x dx \Rightarrow$$

$$\Rightarrow \frac{\ln(3-y)}{-1} = x^2 + c \Rightarrow \ln(3-y)^{-1} = x^2 + c \Rightarrow$$

$$\Rightarrow \boxed{\frac{1}{3-y} = e^{x^2+c}}$$

Lineal de 1° orden

$$y = uv \Rightarrow y' = u'v + uv'$$

$$u'v + uv' + 2xuv = 6x$$

$$u'v + u(\underbrace{v' + 2xv}_{=0}) = 6x \quad (1)$$

$$v' + 2xv = 0 \Rightarrow \frac{dv}{dx} = -2xv \Rightarrow$$

$$\Rightarrow \int \frac{dv}{v} = \int -2x dx \Rightarrow \ln v = -x^2$$

$$\Rightarrow v = e^{-x^2}$$

Reemplazando en (1):

$$u'e^{-x^2} = 6x \Rightarrow \frac{du}{dx} = 6xe^{x^2} \Rightarrow \int du = \int 6xe^{x^2} dx$$

$$\Rightarrow u = 3(e^{x^2} + c)$$

$$y = uv \Rightarrow y = 3(e^{x^2} + c)e^{-x^2} = 3 + 3ce^{-x^2}$$

$$y = 3 + 3ce^{-x^2} \Rightarrow 3-y = -3ce^{-x^2} \Rightarrow 3-y = e^{\ln(-3c)} e^{-x^2}$$

$$\Rightarrow 3-y = e^{-x^2-q} \Rightarrow \frac{1}{3-y} = \frac{1}{e^{-x^2-q}} \Rightarrow \boxed{\frac{1}{3-y} = e^{x^2+q}}$$

$$15) \quad y' + \frac{y}{x+1} = \tan x \quad y = uv \Rightarrow y' = u'v + uv'$$

$$u'v + uv' + \frac{uv}{x+1} = \tan x$$

$$u'v + u \left(v' + \frac{v}{x+1} \right) = \tan x \quad (1)$$

$$v' + \frac{v}{x+1} = 0 \Rightarrow \frac{dv}{dx} = -\frac{v}{x+1} \Rightarrow \int \frac{dv}{v} = \int -\frac{dx}{x+1} \Rightarrow$$

$$\ln v = -\ln(x+1) \Rightarrow v = \frac{1}{x+1}$$

$$\text{Reinsub en (1)} \rightarrow u' \frac{1}{x+1} = \tan x \Rightarrow$$

$$\Rightarrow u' = x \tan x + \tan x \Rightarrow \frac{du}{dx} = x \tan x + \tan x \Rightarrow$$

$$\Rightarrow \int du = \int (x \tan x + \tan x) dx \Rightarrow$$

$$\Rightarrow u = \tan x - x \cos x - \cos x + C$$

$$y = uv \Rightarrow y = \frac{\tan x - x \cos x - \cos x + C}{x+1}$$

$$\text{Passe par } \left(\frac{\pi}{2}, 1\right) \Rightarrow 1 = \frac{\tan \frac{\pi}{2} - \frac{\pi}{2} \cos \frac{\pi}{2} - \cos \frac{\pi}{2} + C}{\frac{\pi}{2} + 1} \Rightarrow$$

$$\Rightarrow \frac{\pi}{2} + 1 = 1 + C \Rightarrow C = \frac{\pi}{2} \Rightarrow y = \frac{\tan x - x \cos x - \cos x + \frac{\pi}{2}}{x+1}$$

$$16) \quad y' + y = 2 \operatorname{sen} x \quad \text{tal que} \quad y(0) = 1$$

$$y = uv \Rightarrow y' = u'v + uv'$$

$$u'v + u v' + uv = 2 \operatorname{sen} x \quad (1)$$

$$u'v + u(v' + v) = 2 \operatorname{sen} x$$

$$v' + v = 0 \Rightarrow \frac{dv}{dx} = -v \Rightarrow \int \frac{dv}{v} = \int -dx \Rightarrow$$

$$\Rightarrow \ln v = -x \Rightarrow v = e^{-x}$$

$$\text{Reemplazando en (1)} \Rightarrow u' e^{-x} = 2 \operatorname{sen} x \Rightarrow$$

$$\Rightarrow \frac{du}{dx} = 2 e^x \operatorname{sen} x \Rightarrow \int du = \int 2 e^x \operatorname{sen} x \Rightarrow$$

$$\Rightarrow u = \frac{2}{1+1} \left[\frac{e^x (\operatorname{sen} x - \cos x)}{1+1} \right] + C$$

$$y = uv \Rightarrow y = \left[e^x (\operatorname{sen} x - \cos x) + C \right] e^{-x}$$

$$y = \operatorname{sen} x - \cos x + C e^{-x}$$

$$\text{Pasa por } (0, 1) \rightarrow 1 = \cancel{\operatorname{sen} 0}^0 - \cos 0 + C e^0$$

$$1 = -1 + C \Rightarrow C = 2$$

$$\boxed{y = \operatorname{sen} x - \cos x + 2e^{-x}}$$

$$18) \frac{dT}{dt} = -k(T - T_A) \quad k > 0$$

para $t=0 \Rightarrow T_0 = T(0) > T_A$

$$a) \frac{dT}{dt} = -k(T - T_A) \Rightarrow \frac{dT}{T - T_A} = -k dt \Rightarrow$$

$$\Rightarrow \ln(T - T_A) = -kt + c \Rightarrow T - T_A = e^{(-kt+c)}$$

$$\text{Siendo } T(0) = T_0: \quad T_0 - T_A = e^{0+c} \Rightarrow e^c = T_0 - T_A$$

$$\Rightarrow c = \ln(T_0 - T_A)$$

$$T - T_A = e^{-kt} e^{\ln(T_0 - T_A)} = e^{-kt} (T_0 - T_A)$$

$$T - T_A = e^{-kt} (T_0 - T_A)$$

$$\boxed{T = e^{-kt} (T_0 - T_A) + T_A}$$

con $k > 0$

$$b) \lim_{t \rightarrow +\infty} T(t) = \lim_{t \rightarrow +\infty} [e^{-kt} (T_0 - T_A) + T_A] = T_A$$

30 minutos $\rightarrow 0,5$ hora

$$c) T_A = 0 \quad t = 30 \quad T(30) = \frac{T_0}{2}$$

$$T\left(\frac{1}{2}\right) = e^{-k \cdot 1/2} (T_0 - 0) + 0 = \frac{T_0}{2} \Rightarrow$$

$$\Rightarrow e^{-k/2} \cancel{T_0} = \frac{\cancel{T_0}}{2} \Rightarrow e^{-k/2} = \frac{1}{2} \Rightarrow$$

$$\Rightarrow -\frac{k}{2} = \ln \frac{1}{2} \Rightarrow -\frac{k}{2} = \ln 2^{-1} \Rightarrow \frac{k}{2} = \ln 2 \Rightarrow$$

$$\Rightarrow \boxed{k = 2 \ln 2}$$

$$19) a) x^2 + 4y^2 = C_1 \Rightarrow 2x + 8y y' = 0 \Rightarrow y' = -\frac{2x}{8y} \Rightarrow y' = -\frac{x}{4y}$$

$$b) y = C_2 x^4 \Rightarrow y' = 4C_2 x^3 \Rightarrow y' = 4 \frac{y}{x} \Rightarrow y' = 4 \frac{y}{x}$$

$$\left. \begin{array}{l} y'_a = -\frac{x}{4y} \\ y'_b = \frac{4y}{x} \end{array} \right\} \Rightarrow y'_a = -\frac{1}{y'_b}$$

$$19) b) a) x^2 - y^2 + \ln(\cos 2xy) = C_1 \Rightarrow 2x - 2y y' + \frac{-\sin 2xy \cdot 2(y + y'x)}{\cos 2xy} = 0$$

$$\Rightarrow x - y y' + \tan 2xy \cdot y - \tan 2xy \cdot y' x = 0 \Rightarrow$$

$$\Rightarrow x - y \operatorname{tg} 2xy = y y' + y' x \operatorname{tg} 2xy \Rightarrow$$

$$\Rightarrow x - y \operatorname{tg} 2xy = y' (y + x \operatorname{tg} 2xy) \Rightarrow$$

$$\Rightarrow y' = \frac{x - y \operatorname{tg} 2xy}{y + x \operatorname{tg} 2xy}$$

$$b) x^2 - y^2 + \ln \sin(2xy) = c \Rightarrow \cancel{2x} - \cancel{2y} y' + \frac{2 \cos 2xy}{\sin 2xy} (y + y' x) = 0$$

$$\Rightarrow x - y y' + \operatorname{ctg} 2xy (y + y' x) = 0 \Rightarrow$$

$$\Rightarrow x - y y' + y \operatorname{ctg} 2xy + y' x \operatorname{ctg} 2xy = 0 \Rightarrow$$

$$\Rightarrow x + y \operatorname{ctg} 2xy = y y' - y' x \operatorname{ctg} 2xy \Rightarrow$$

$$\Rightarrow \frac{x + y \operatorname{ctg} 2xy}{y - x \operatorname{ctg} 2xy} = y'$$

$$- \frac{1}{y'} = - \frac{y + x \operatorname{tg} 2xy}{x - y \operatorname{tg} 2xy} = \frac{y + x \operatorname{tg} 2xy}{-x + y \operatorname{tg} 2xy} =$$

$$= \frac{\operatorname{ctg} 2xy}{\operatorname{tg} 2xy} \frac{y + x \operatorname{tg} 2xy}{y \operatorname{tg} 2xy - x} = \frac{y \operatorname{ctg} 2xy + x}{y - x \operatorname{ctg} 2xy} =$$

$$= \frac{x + y \operatorname{ctg} 2xy}{y - x \operatorname{ctg} 2xy} = y'$$

20

a) $y' = 2$

Hacemos $-\frac{1}{y'} = 2 \Rightarrow \frac{dx}{dy} = -2 \Rightarrow \int dx = \int -2 dy \Rightarrow$

$$\Rightarrow x = -2y + c \Rightarrow y = \frac{x - c}{-2} \Rightarrow \boxed{y = k - \frac{x}{2}}$$

$$20) b) \quad y = c e^x \Rightarrow c = y e^{-x}$$

$$y' = c e^x \Rightarrow y' = y e^{-x} e^x = y$$

$$y' = y \quad \text{ontog} \rightarrow -\frac{1}{y'} = y \Rightarrow -\frac{1}{y} = y' \Rightarrow -\frac{1}{y} = \frac{dy}{dx} \rightarrow$$

$$\Rightarrow \int dx = \int \frac{1}{y} dy \Rightarrow -x + c = \frac{y^2}{2} \Rightarrow \frac{y^2}{2} + x = c$$

$$20) c) \quad y(c x + 1) = x \rightarrow c x + 1 = \frac{x}{y} \Rightarrow c = \left(\frac{x}{y} - 1 \right) \frac{1}{x}$$

$$y'(c x + 1) + y c = 1$$

$$y' \frac{x}{y} + y \left(\frac{x}{y} - 1 \right) \frac{1}{x} = 1$$

$$y' \frac{x}{y} + (x - y) \frac{1}{x} = 1$$

$$y' \frac{x}{y} + \cancel{x} - \frac{y}{x} = \cancel{x} \Rightarrow y' \frac{x}{y} = \frac{y}{x} \Rightarrow$$

$$\Rightarrow y' x^2 = y^2 \quad \xrightarrow{\perp} \quad -\frac{1}{y'} x^2 = y^2 \Rightarrow$$

$$\Rightarrow -\frac{dx}{dy} x^2 = y^2 \Rightarrow -\int x^2 dx = \int y^2 dy \Rightarrow$$

$$\Rightarrow -\frac{x^3}{3} + c = \frac{y^3}{3} \Rightarrow \boxed{y^3 = -x^3 + a}$$

$$2o) 1) y = \ln(x+c) \quad (1)$$

1º) hallamos la ED del haz original

$$y' = \frac{1}{x+c} \quad \left\{ \begin{array}{l} \Rightarrow y' = \frac{1}{e^y} \Rightarrow y' = e^{-y} \end{array} \right.$$

De (1): $x+c = e^y$

2º) sustituimos y' por $-\frac{1}{y'}$ en la ED hallada para obtener la ED del haz ortogonal

$$-\frac{1}{y'} = e^{-y}$$

3º) Resolvemos la nueva ED

$$-\frac{dx}{dy} = e^{-y} \Rightarrow -dx = e^{-y} dy \Rightarrow$$

$$\Rightarrow \int -dx = \int e^{-y} dy \Rightarrow -x+c = \frac{e^{-y}}{-1} \Rightarrow$$

$$\Rightarrow e^{-y} = x-c = x+k$$

$$\Rightarrow -y = \ln(x+k) \Rightarrow \boxed{y = -\ln(x+k)}$$

La SG obtenida es la ecuación de haz de curvas ortogonal al dado

20) e) $y = C \operatorname{sen}(2x)$ ①

1°) Hallamos la ED asociada al haz de curvas original

$$\left. \begin{array}{l} y' = 2C \cos(2x) \\ \text{De ①: } C = \frac{y}{\operatorname{sen}(2x)} \end{array} \right] \Rightarrow y' = 2 \frac{y}{\operatorname{sen}(2x)} \cos(2x)$$

$$\Rightarrow y' = \frac{2y}{\operatorname{tg}(2x)} \quad \text{ED asociada al haz original}$$

2°) En la ED anterior reemplazamos y' por $-\frac{1}{y'}$ para obtener la ED del haz ortogonal: $-\frac{1}{y'} = \frac{2y}{\operatorname{tg}(2x)}$

3°) Resolvemos la nueva ED

$$-\frac{1}{y'} = \frac{2y}{\operatorname{tg}(2x)} \Rightarrow -\frac{1}{\frac{dy}{dx}} = \frac{2y}{\operatorname{tg}(2x)} \Rightarrow$$

$$\Rightarrow -\frac{dx}{dy} = \frac{2y}{\operatorname{tg}(2x)} \Rightarrow -\operatorname{tg}(2x) dx = 2y dy$$

$$\int -\operatorname{tg}(2x) dx = \int 2y dy \Rightarrow$$

$$\Rightarrow \boxed{\frac{1}{2} \ln |\cos(2x)| = y^2 + C}$$

La SG hallada es el haz de curvas ortogonal que se busca

$$f) (x+y)^2 = kx^2 \quad k > 0 \rightarrow k = \left(\frac{x+y}{x}\right)^2$$

$$\cancel{2}(x+y) \cdot (1+y') = \cancel{2}kx \Rightarrow \cancel{(x+y)}(1+y') = \frac{(x+y)^2}{x^2}$$

$$(1+y') = \frac{x+y}{x} \xrightarrow{1} 1 - \frac{1}{y'} = \frac{x+y}{x} \Rightarrow$$

$$\Rightarrow -\frac{1}{y'} = \frac{x+y}{x} - 1 = \frac{\cancel{x+y} - x}{x} \Rightarrow$$

$$\Rightarrow -\frac{1}{y'} = \frac{y}{x} \Rightarrow -\frac{dx}{dy} = \frac{y}{x} \Rightarrow -\int dx \cdot x = \int y \, dy \Rightarrow$$

$$\Rightarrow -\frac{x^2}{2} + c = \frac{y^2}{2} \Rightarrow \frac{y^2}{2} + \frac{x^2}{2} = c \Rightarrow \boxed{y^2 + x^2 = a}$$

21)

$$\bullet y^3 = \Delta x \Rightarrow \Delta = \frac{y^3}{x}$$

$$3y^2 y' = \Delta \Rightarrow \cancel{3}y^2 y' = \frac{y^3}{x} \Rightarrow y' = \frac{y}{3x}$$

$$\bullet x^2 + ay^2 = B^2$$

$$\cancel{2}x + \cancel{2}ay' y = 0 \Rightarrow y' = -\frac{x}{ay}$$

$$\left. \begin{aligned} -\frac{1}{y'} &= \frac{ay}{x} \\ y'a &= \frac{y}{3x} \end{aligned} \right\} \Rightarrow \frac{ay}{x} = \frac{y}{3x} \Rightarrow a = \frac{1}{3} \Rightarrow$$

$$\Rightarrow \boxed{a = \frac{1}{3}}$$

$$22) \quad \left. \begin{aligned} y &= cx + c \\ y' &= c \end{aligned} \right\} \Rightarrow \left. \begin{aligned} c(x+1) &= y \\ y' &= c \end{aligned} \right\} \Rightarrow y' = \frac{y}{x+1}$$

linha $\perp \Rightarrow -\frac{1}{y'} = \frac{y}{x+1} \Rightarrow -x-1 = yy' = 0$

$$\Rightarrow -x-1 = y \frac{dy}{dx} \Rightarrow \int (-x-1) dx = \int y dy \Rightarrow$$

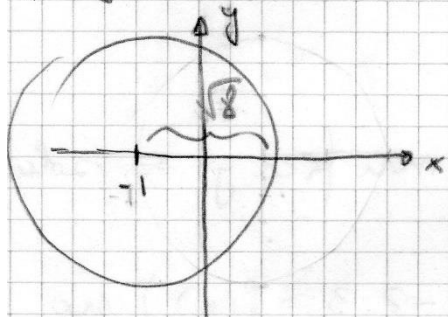
$$\Rightarrow -\frac{x^2}{2} - x + k = \frac{y^2}{2} \Rightarrow \frac{y^2}{2} + \frac{x^2}{2} + x = c$$

Pase por $(1,2) \rightarrow \frac{4^2}{2} + \frac{1^2}{2} + 1 = c \Rightarrow c = \frac{7}{2}$

Curva $\rightarrow \frac{y^2}{2} + \frac{x^2}{2} + x = \frac{7}{2}$

$$y^2 + x^2 + 2x = 7$$

$$y^2 + (x^2 + 2x + 1) = 7+1 \Rightarrow y^2 + (x+1)^2 = 8$$



Círculo centro en $(-1, 0)$

y radio $\sqrt{8}$

$$\text{Long círculo} = 2\pi \cdot r = 2\pi \sqrt{8} =$$

$$= 2\pi \cdot 2\sqrt{2} =$$

$$= \boxed{4\pi\sqrt{2}}$$

$$23) \quad \begin{cases} x y'' - 2y' = 0 \\ w = y' \end{cases} \Rightarrow x w' - 2w = 0 \Rightarrow$$

$$\Rightarrow w' - \frac{2}{x} w = 0 \Rightarrow w' = \frac{2}{x} w \Rightarrow$$

$$\Rightarrow \frac{dw}{dx} = \frac{2}{x} w \Rightarrow \int \frac{dw}{w} = \int \frac{2}{x} dx \Rightarrow$$

$$\Rightarrow \ln kw = 2 \ln x \Rightarrow \ln kw = \ln x^2 \Rightarrow$$

$$\Rightarrow kw = x^2$$

$$y'(1) = 3 \Rightarrow w(1) = 3 \quad \text{luego } k \cdot 3 = 1$$

$$\Rightarrow k = 1/3$$

$$\frac{w}{3} = x^2 \quad \therefore \quad \frac{y'}{3} = x^2 \Rightarrow \frac{dy}{dx} = 3x^2 =$$

$$\Rightarrow \int dy = \int 3x^2 dx \Rightarrow y = x^3 + C$$

$$y(1) = 3 \Rightarrow 3 = 1^3 + C \Rightarrow C = 3 - 1 = 2$$

$$\boxed{y = x^3 + 2}$$

$$24) \quad y'' - 2y' = x \quad w = y'$$

$$w' - 2w = x$$

$$\text{Sea } w = ur \Rightarrow w' = u'r + ur'$$

$$u'r + ur' - 2ur = x$$

$$u'r + u(r' - 2r) = x$$

$$r' - 2r = 0 \Rightarrow \frac{dr}{dx} = 2r \Rightarrow \int \frac{dr}{r} = \int 2 dx \Rightarrow$$

$$\Rightarrow \ln r = 2x \Rightarrow r = e^{2x}$$

$$u' e^{2x} = x \Rightarrow \frac{du}{dx} = x e^{-2x} \Rightarrow \int du = \int x e^{-2x} dx \Rightarrow$$

$$\Rightarrow u = \frac{e^{-2x}}{4} (-2x-1) + c$$

$$w = ur \Rightarrow w = \left[\frac{e^{-2x}}{4} (-2x-1) + c \right] e^{2x}$$

$$w = \frac{-2x-1}{4} + ce^{2x}$$

$$y' = \frac{-2x-1}{4} + ce^{2x}$$

$$\int dy = \int \left[\frac{-2x-1}{4} + ce^{2x} \right] dx$$

$$y = -\frac{1}{2} \frac{x^2}{2} - \frac{1}{4} x + \frac{c e^{2x}}{2} + k$$

$$y = -\frac{x^2}{4} - \frac{x}{4} + \frac{c}{2} e^{2x} + k$$

$$25) \quad y''' - y'' = 0 ; \text{ let } w = y''$$

$$w' - w = 0 \Rightarrow \frac{dw}{dx} = w \Rightarrow \int \frac{dw}{w} = \int \frac{dx}{1} \Rightarrow$$

$$\Rightarrow \ln k w = x \Rightarrow e^x = k w \Rightarrow w = \frac{1}{k} e^x$$

$$y' = c_1 e^x ; \text{ let } r = y'$$

$$r' = c_1 e^x \Rightarrow \frac{dr}{dx} = c_1 e^x \Rightarrow \int dr = \int c_1 e^x dx \Rightarrow$$

$$\Rightarrow r = c_1 e^x + c_2$$

$$y' = c_1 e^x + c_2 \Rightarrow \frac{dy}{dx} = c_1 e^x + c_2 \Rightarrow$$

$$\Rightarrow \int dy = \int (c_1 e^x + c_2) dx \Rightarrow \boxed{y = c_1 e^x + c_2 x + c_3}$$

$$26) \quad x^2 y'' - 2y = f(x)$$

$$y'' - \frac{2y}{x^2} = \frac{f(x)}{x^2} \Rightarrow y'' = \frac{f(x)}{x^2} + \frac{2y}{x^2} \quad (*)$$

$$x y'' - 2y' = f(x)$$

Sea $y = x y_p$

$$y' = y_p + y_p' x$$

$$y'' = y_p' + y_p'' x + y_p'$$

$$x (y_p' + y_p'' x + y_p') - 2(y_p + y_p' x) = f(x)$$

$$\cancel{x y_p'} + y_p'' x^2 + \cancel{y_p' x} - 2y_p - \cancel{2y_p' x} = f(x)$$

$$y_p'' x^2 - 2y_p = f(x)$$

reempl en (*)

$$\left(\frac{f(x)}{x^2} + \frac{2y_p}{x^2} \right) x^2 - 2y_p = f(x)$$

$$f(x) + 2y_p - 2y_p = f(x)$$

$$f(x) = f(x)$$

verifico y es soluc

$y = x y_p$ que pasa por $(2, y_0)$. Luego: $y = 2 \cdot 3 = 6$

$$27) \quad x(x+y') = y$$

$$y'x = y - x^2 \Rightarrow y'x - y + x^2 = 0 \Rightarrow y' - \frac{1}{x}y + x = 0$$

es lineal de 1º orden

$$y = ur \Rightarrow y' = u'r + r'u$$

$$u'r + r'u - \frac{ur}{x} + x = 0$$

$$u'r + u(r' - \frac{r}{x}) + x = 0 \quad (1)$$

$$r' - \frac{r}{x} = 0 \Rightarrow \frac{dr}{dx} = \frac{r}{x} \Rightarrow \int \frac{dr}{r} = \int \frac{dx}{x} \Rightarrow \ln r = \ln x$$

$$\Rightarrow r = x$$

lemplo en ①

$$u'x + x = 0 \Rightarrow u'x = -x \Rightarrow u' = -1 \Rightarrow \frac{du}{dx} = -1 \Rightarrow$$

$$\Rightarrow \int du = \int dx \Rightarrow u = -x + C$$

$$y = ur \Rightarrow y = (-x + C)x \Rightarrow y = -x^2 + Cx$$

$$\text{base por } (a, 0) \Rightarrow 0 = -a^2 + C \cdot a \Rightarrow C = \frac{a^2}{a} = a \quad (a > 0)$$

$$\text{luego SP} \rightarrow y = -x^2 + ax$$

$$y' = -2x + a = 0 \Rightarrow x = \frac{a}{2} \quad \text{pto crítico}$$

$$y'' = -2 < 0 \Rightarrow y''\left(\frac{a}{2}\right) = -2 < 0 \rightarrow \text{máx relat.}$$

Re ser $f\left(\frac{a}{2}\right) = y\left(\frac{a}{2}\right) = -\left(\frac{a}{2}\right)^2 + a\left(\frac{a}{2}\right)$ el único
máx de la f en $(0, a)$, debe evaluarse en los
extremos

$$\left. \begin{array}{l} y\left(\frac{a}{2}\right) = -\frac{a^2}{4} + \frac{a^2}{2} = \frac{-a^2 + 2a^2}{4} = \frac{a^2}{4} > 0 \\ y(0) = 0 \\ y(a) = -a^2 + a^2 = 0 \end{array} \right\} \text{luego} \Rightarrow$$

$\Rightarrow f\left(\frac{a}{2}\right)$ es el máx valor que alcanza f en
 $[0, a]$ y es su máx absoluto en dicho intervalo
 $\frac{a^2}{4}$ es el máx abs.

$$28) \quad a) \quad (x^2-1)y' + xy(1-y) = 0$$

$$y' = \frac{xy(y-1)}{x^2-1}$$

$$\frac{dy}{dx} = \frac{xy(y-1)}{x^2-1} \Rightarrow \int \frac{dy}{y(y-1)} = \int dx \frac{x}{x^2-1} \Rightarrow 0$$

$$\Rightarrow \frac{1}{-1} \ln\left(\frac{y}{y-1}\right) = \frac{1}{2} \ln(x^2-1) + \ln k$$

$$\Rightarrow \ln\left(\frac{y}{y-1}\right)^{-1} = \ln[(x^2-1)^{1/2} \cdot k] \Rightarrow 0$$

$$\Rightarrow \boxed{\frac{y-1}{y} = k \sqrt{x^2-1}} \rightarrow y-1 = k \cdot y \sqrt{x^2-1} \Rightarrow (y-1)^2 = k^2 y^2 (x^2-1)$$

$$b) \quad y=2 \wedge x=3 \text{ única solución}$$

$$\frac{2-1}{2} = k \sqrt{9-1} \Rightarrow \frac{1}{2} = k \sqrt{8} \Rightarrow k = \frac{1}{2\sqrt{8}}$$

$$\boxed{\frac{y-1}{y} = \frac{1}{2\sqrt{8}} \sqrt{x^2-1}}$$

$$(2\sqrt{8})^2 = 4 \cdot 8 = 32$$

$$c) \quad y=1 \wedge x=-1$$

$$\frac{1-1}{1} = k \sqrt{1-1} \Rightarrow 0=0 \text{ vale } \forall k \rightarrow \text{infinitas soluciones}$$

$$y=1 \wedge x=1$$

$$\frac{1-1}{1} = k \sqrt{1-1} \Rightarrow 0=0 \text{ vale } \forall k \rightarrow \text{infinitas soluciones}$$

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29) $y = x^3 - x$ es soluc de $y' + \beta(x)y = 2x^2$

$y' = 3x^2 - 1$ y verifica la ED, luego

$$(3x^2 - 1) + \beta(x)(x^3 - x) = 2x^2 \Rightarrow \beta(x) = \frac{2x^2 + 1 - 3x^2}{x^3 - x} \Rightarrow$$

$$\Rightarrow \beta(x) = \frac{-x^2 + 1}{x^3 - x} = \frac{1 - x^2}{-x(1 - x^2)} \Rightarrow \beta(x) = -\frac{1}{x}$$

$$y' + \left(-\frac{1}{x}\right)y = 2x^2 \quad \text{lineal 1º orden} \quad \begin{cases} y = uv \\ y' = u'v + uv' \end{cases}$$

$$u'v + uv' - \frac{1}{x}uv = 2x^2$$

$$u'v + u \underbrace{\left(v' - \frac{1}{x}v\right)}_{=0} = 2x^2 \quad (1)$$

$$v' - \frac{1}{x}v = 0 \Rightarrow \frac{dv}{dx} = \frac{v}{x} \Rightarrow \int \frac{dv}{v} = \int \frac{dx}{x} \Rightarrow$$

$$\Rightarrow \ln v = \ln x \Rightarrow v = x$$

Reemplazando en (1):

$$u'x = 2x^2 \Rightarrow u' = 2x \Rightarrow \frac{du}{dx} = 2x \Rightarrow$$

$$\Rightarrow \int du = \int 2x dx \Rightarrow u = x^2 + c$$

$$\text{Luego } y = uv \Rightarrow y = (x^2 + c)x \Rightarrow y = x^3 + cx$$

La SP verifica que $y(1) = 4$, entonces

$$4 = 1^3 + c \Rightarrow c = 3$$

$$SP \rightarrow \boxed{y = x^3 + 3x}$$